

NAME : _____

CLASS : _____



JURONG PIONEER JUNIOR COLLEGE
JC2 Preliminary Examination 2022

BIOLOGY
Higher 2

9744/01
23 September 2022

Paper 1 Multiple Choice

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, class and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of **21** printed pages and **1** blank page.

1 Which cell structures contain rRNA?

- 1 chloroplast
- 2 mitochondrion
- 3 nucleus
- 4 smooth endoplasmic reticulum

- A** 1, 2, 3 and 4
- B** 1, 2 and 3 only
- C** 1, 2 and 4 only
- D** 3 and 4 only

2 It is now possible to detect and follow the movement of single fluorescent molecules in living cells. For example, tracking the movement of fluorescent molecules of RNA polymerase in the bacterium *Escherichia coli* shows that some of these molecules move randomly through the cytoplasm (mobile molecules) but others remain in one place (immobile molecules).

Which statement explains one of these observations in *E. coli* about fluorescent RNA polymerase?

- A** The immobile molecules are attached to DNA and are transcribing DNA into mRNA in the cytoplasm.
- B** The immobile molecules are trapped inside the nuclear envelope and are transcribing DNA into mRNA.
- C** The mobile molecules are attached to 70S ribosomes and are translating mRNA to protein.
- D** The mobile molecules are attached to 80S ribosomes that move randomly through the cytoplasm.

3 Which statements about glycosidic bonds are correct?

- 1 They occur by condensation reactions between the OH groups of two molecules.
- 2 They occur only between glucose molecules.
- 3 The bonds can only be formed between carbon 1 and carbon 4 or carbon 1 and carbon 6 of adjacent molecules.
- 4 Hydrolysis of the bonds releases energy.

A 1, 2 and 3

B 1 and 4

C 2 and 4

D 3 and 4

4 Which properties of phospholipids explain why single layers of phospholipids added to water immediately form bilayers?

- 1 The hydrophobic fatty acid chains repel water molecules so that the tails pack together.
- 2 The non-polar fatty acid chains are attracted to each other by hydrophobic interactions.
- 3 Hydrogen bonds form between the phosphate groups and water.

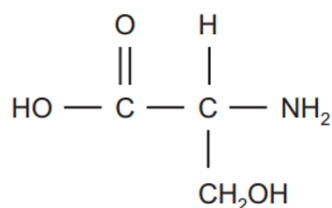
A 1, 2 and 3

B 1 and 2 only

C 1 and 3 only

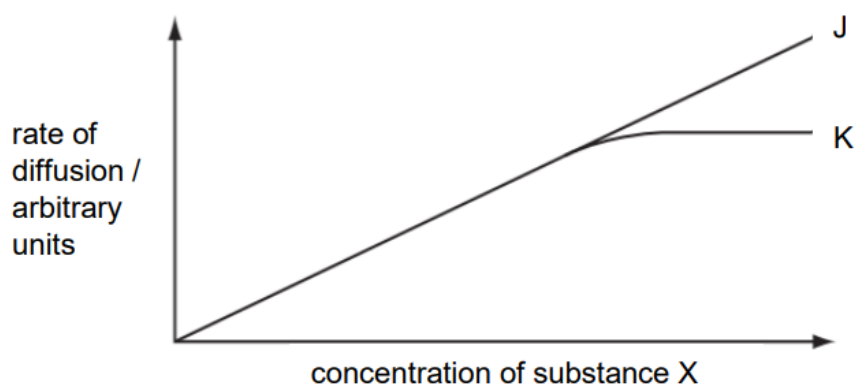
D 2 and 3 only

- 5 The diagram shows a molecule.



Which test on a polymer of this molecule would give a positive result?

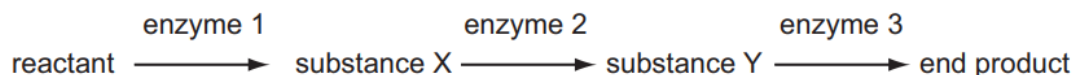
- A** adding biuret solution
- B** adding iodine in potassium iodide solution
- C** heating with Benedict's solution
- D** shaking with ethanol then pouring into water
- 6 The graph shows rates of simple diffusion and facilitated diffusion of substance X across the cell surface membrane.



Which row is correct?

	graph K	explanation
A	simple diffusion	simple diffusion allows the concentrations of substance X inside and outside the cell to be equal in a shorter time
B	facilitated diffusion	facilitated diffusion is limited by the number of protein channels in the membrane
C	simple diffusion	simple diffusion is limited by the number of transient gaps in the membrane
D	facilitated diffusion	facilitated diffusion is limited by the amount of ATP available

- 7 A metabolic pathway is shown here.



What would be the effect of adding a small amount of a non-competitive inhibitor of enzyme 2?

- A** Enzyme 2 would be denatured.
- B** Substance X would increase in concentration.
- C** Substance Y would no longer be formed.
- D** The initial reactant would no longer be metabolised.
- 8 Biomolecules that are specific to particular types of cell, including stem cells, are known as cell markers. These markers can be proteins, glycoproteins or glycolipids. Some examples are listed.
- Zygotic stem cells are positive for ZScan4 and OCT4 but negative for CD34 and CD45.
 - Embryonic stem cells at the 32-celled stage are positive for OCT4 and negative for ZScan4, CD34 and CD45.
 - Blood stem cells that give rise to white blood cells are positive for CD34 and negative for ZScan4 and OCT4.
 - In later developmental stages, some blood stem cells become positive for CD45.
 - Differentiated blood cells are negative for ZScan4, OCT4 and CD34 but most types are positive for CD45.

Which statement is consistent with this information?

- A** Cells with the CD34 cell marker are multipotent.
- B** Cells with the CD45 cell marker are multipotent.
- C** Cells with the OCT4 cell marker are totipotent.
- D** Cells with the ZScan4 cell marker are pluripotent.

- 9** A piece of a DNA molecule contains 84 base pairs. The table shows the number of adenine and cytosine bases in one or both of the DNA strands in this piece of DNA molecule.

base	strand 1	strand 2
adenine	28	23
cytosine	15	

How many guanine bases are present in this piece of DNA molecule?

- A** 18
- B** 33
- C** 36
- D** 41
- 10** In 2009, the H1N1 influenza outbreak caused nearly 15 000 deaths worldwide. The highly virulent virus was formed by antigenic shift.

Which statement is most likely to have resulted in antigenic shift?

- A** Chance mutations occurring in a strain of influenza, giving rise to novel haemagglutinin proteins.
- B** Recombination of viral genes within a host cell during infection.
- C** Simultaneous infection of a cell by two or more strains of influenza.
- D** High error rate in influenza RNA-dependent RNA polymerase resulting in new strains upon viral reproduction.

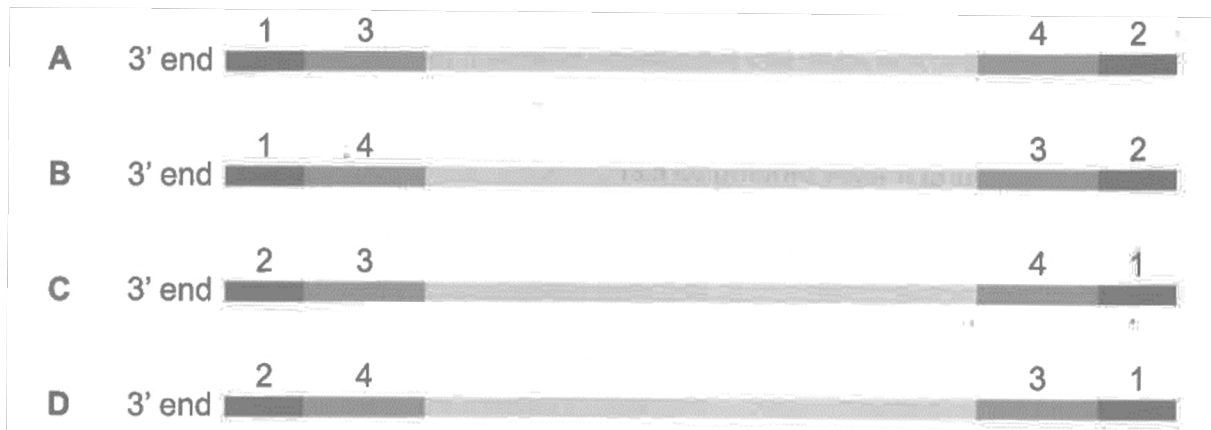
11 Which row correctly describes the transfer of DNA from one bacterium to another?

	binary fission	transduction	conjugation
A	bacterial chromosome and plasmids pass to daughter cells	DNA transferred by bacteriophage from one bacterium to another	single strand of F plasmid transferred from one bacterium to another
B	bacterial chromosome and plasmids pass to daughter cells	bacterial cell takes up foreign DNA from culture medium	double strand of F plasmid transferred from one bacterium to another
C	only plasmids pass to daughter cells	DNA transferred by bacteriophage from one bacterium to another	single strand of F plasmid transferred from one bacterium to another
D	only plasmids pass to daughter cells	bacterial cell takes up foreign DNA from culture medium	double strand of F plasmid transferred from one bacterium to another

12 Processing of mRNA in eukaryotes results in a molecule that has four regions, in addition to the translated exon.

- 1 a poly-A tail
- 2 a cap consisting of modified guanine nucleotides
- 3 an untranslated region immediately before the translation start codon
- 4 an untranslated region after the translation termination sequence

Which diagram shows the correct arrangement of these regions in the processed mRNA molecule?



13 During agarose gel electrophoresis, which factors affect the rate of migration of DNA fragments through the gel?

- 1 density of loading dye
- 2 size of agarose gel
- 3 size of DNA fragments
- 4 voltage used

- A** 1, 2 and 3
B 1 and 2 only
C 3 and 4
D 3 only

14 How many statements about gene mutations are correct?

- 1 Gene mutations are always inherited and never acquired during an individual's lifetime.
- 2 Gene mutations always give rise to recessive alleles.
- 3 Gene mutations may have no effect on the encoded proteins.
- 4 Gene mutations may result in new phenotypes.

- A** 1
B 2
C 3
D 4

- 15 The diagram shows the chromosomes of a typical plant cell at the metaphase stage of mitosis.



Which row describes this cell during metaphase?

	diploid number (2n) for the plant	structures present at metaphase		
		cell wall	centriole	spindle
A	4	✓	×	✓
B	8	×	✓	✓
C	8	✓	×	✓
D	16	✓	✓	×

- 16 The DNA in cells that are dividing is much more sensitive to damage by ionising radiation than the DNA in cells in interphase. There are mechanisms to repair damaged DNA but if these mechanisms cannot repair the DNA sufficiently, then the cell dies.

Which statements explain why ionising radiation is used to treat cancer?

- 1 Exposure to ionising radiation on its own does not increase the risk of cancer.
- 2 Dysregulation of the checkpoints of cell division is common in cancer cells and leads to uncontrolled cell division.
- 3 Proteins needed for DNA repair are often ineffective in cancer cells due to mutation of their genes.

- A** 1, 2 and 3
B 1 and 2 only
C 1 and 3 only
D 2 and 3 only

17 Which statements describe the significance of meiosis?

- 1 results in haploid cells that fuse to form a cell with a full complement of chromosomes
- 2 contributes to genetic diversity, which is necessary for natural selection
- 3 ensures that offspring will have the same genes as the fitter parent
- 4 improves the relative fitness of a species

- A** 1, 2 and 3
B 1 and 2 only
C 1 and 4
D 2 and 3 only

18 Which statement describes Mendel's law of independent assortment?

- A** Allele pairs separate during gamete formation and recombine during fertilisation.
B Allele pairs for different genes separately independently during gamete formation.
C Unlinked genes are assorted with a 9 : 3 : 3 : 1 ratio in a dihybrid cross.
D Allele pairs for the same gene are assorted independently during gamete formation.

- 19** Retinitis pigmentosa is a rare degenerative disease of the retina that leads eventually to blindness.

Two forms of the disease are as follows:

- Usher syndrome, caused by an autosomal recessive allele, *USH2A*, of a gene on chromosome 1
- a form of the disease affecting rhodopsin, caused by an autosomal dominant allele of the *RHO* gene on chromosome 3.

A man who does not carry *USH2A* has a single copy of the *RHO* mutation. A woman has Usher syndrome.

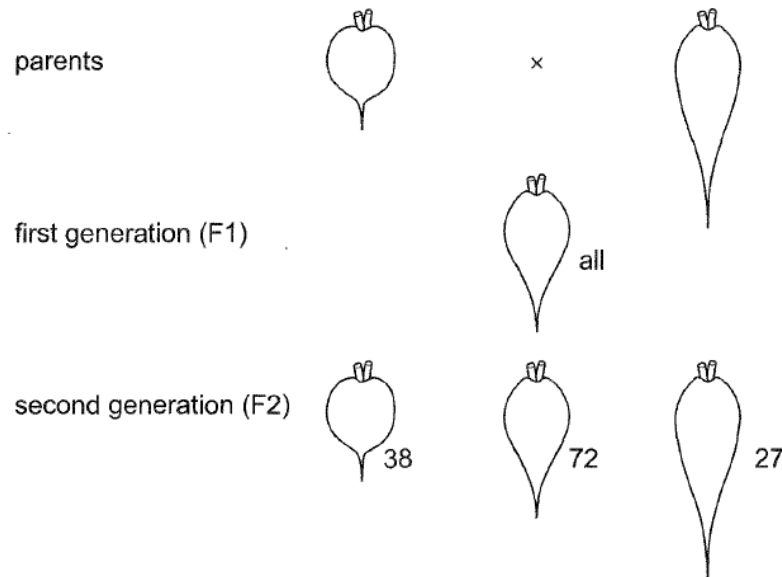
What is the probability of their first child having retinitis pigmentosa?

- A** 0.1875
- B** 0.2500
- C** 0.5000
- D** 0.5625

20 Radishes can have different root shapes.

In 1924, a geneticist tried to work out the genetic basis of this variation. He crossed pure-breeding 'round' and pure-breeding 'conical' radishes to produce an F1 generation. The F1 offspring were then crossed to produce an F2 generation.

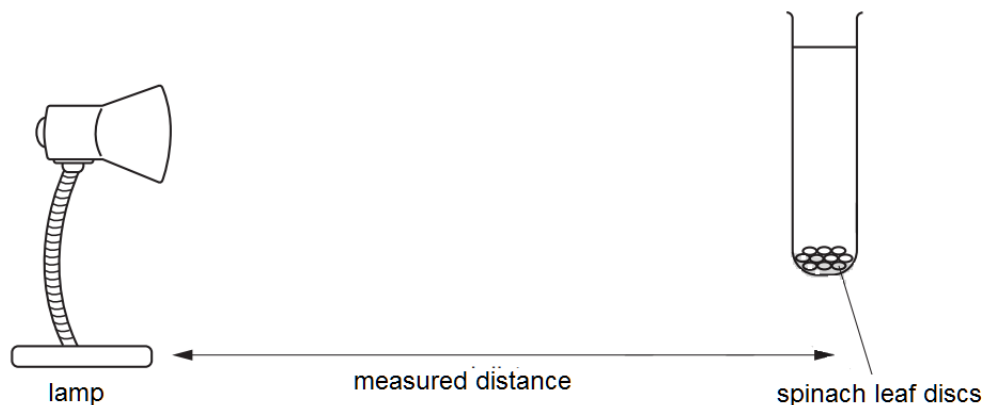
He presented his results as a diagram.



Which conclusion about the variation of root shape could explain his results?

- A** It is an example of continuous variation controlled by three or more genes with additive effects.
- B** It is an example of continuous variation controlled by one or two genes with additive effects.
- C** It is an example of discontinuous variation controlled by one gene with two alleles showing codominance.
- D** It is an example of discontinuous variation controlled by one gene with multiple alleles.

- 21** A student carried out an investigation into the effect of light intensity on photosynthesis. Several groups of spinach leaf discs were placed in test tubes of water. The discs all sank to the bottom of the tubes. Each tube was placed at a measured distance from a lamp.



As photosynthesis occurs, the build-up of oxygen gas in the leaf discs causes them to rise from the bottom of the tubes upwards. The results are shown in the table below.

tube number	distance from lamp / mm	time taken for five discs to float / s
1	50	125
2	100	210
3	150	360
4	200	600
5	250	none floated in the time available

Which of these statements are true?

- 1 The compensation point occurs between 200 and 250 mm.
- 2 A variable which is controlled is the distance of the tube from the light source.
- 3 The time taken for the discs to rise is directly proportional to the distance from the lamp.

- A** 1, 2 and 3
- B** 1 and 3 only
- C** 1 only
- D** 2 and 3 only

22 Which statements apply both to respiration in aerobic conditions and to respiration in anaerobic conditions in the muscle cells of a mammal?

- 1 They both lead to the release of CO₂.
- 2 They both result in the production of ATP.
- 3 They both begin in the mitochondrial matrix.
- 4 They both involve the reduction of NAD.

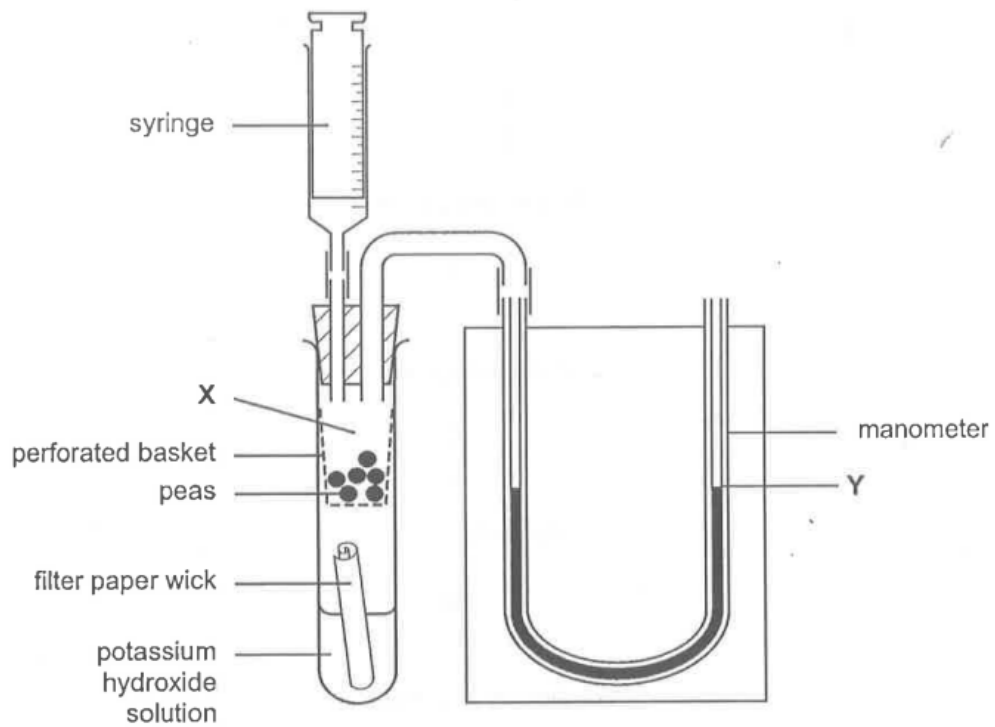
A 1 and 2

B 1 and 3

C 2 and 4

D 3 and 4

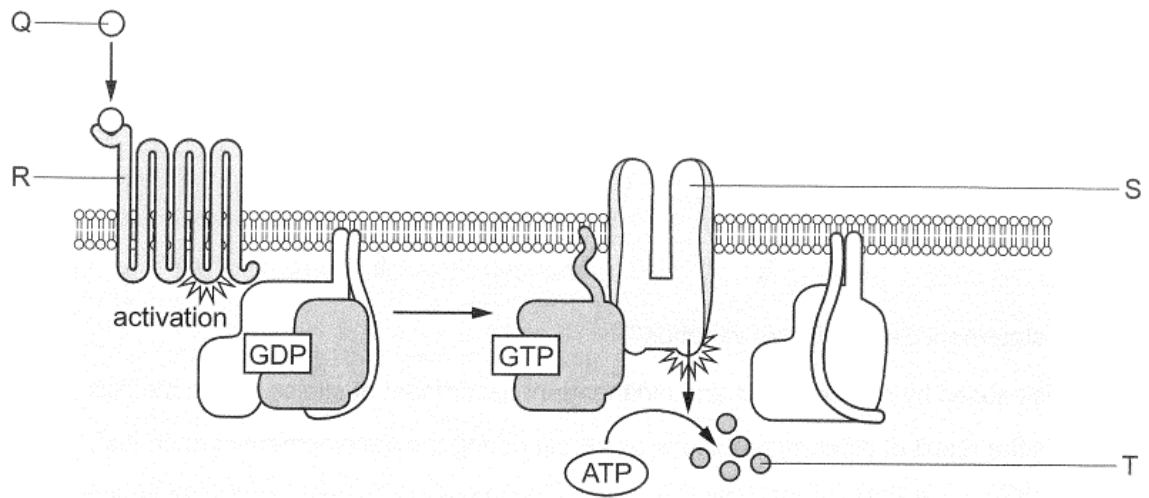
- 23** The diagram shows a respirometer that is used to measure gaseous exchange in peas. The set-up is maintained at a constant temperature.



What happens to the gas pressure at **X** and to the fluid level at **Y**?

	gas pressure at X	fluid level at Y
A	falls	falls
B	falls	rises
C	rises	rises
D	rises	falls

24 The diagram shows part of a cell signalling pathway in a liver cell.



Which row correctly identifies the molecules labelled in the diagram?

	Q	R	S	T
A	glucagon	G-protein linked receptor	adenyl cyclase	cyclic AMP
B	glucagon	G-protein linked receptor	G-protein	ADP
C	insulin	tyrosine kinase receptor	phosphatase	cyclic AMP
D	insulin	tyrosine kinase receptor	adenyl cyclase	ADP

- 25** Zebras with some horizontal stripes are bitten less frequently by tsetse flies than those with only vertical stripes. These flies carry diseases that infect zebras.

Which combination of statements could explain an increase in the proportion of individuals with horizontal stripes in a population of zebras over several generations?

- 1 Tsetse flies are a selection pressure.
- 2 Compared with other zebras, zebras with some horizontal stripes are likely to get fewer diseases from tsetse flies.
- 3 On average, zebras with some horizontal stripes live longer as they are more resistant to disease.
- 4 Zebras with some horizontal stripes pass the allele responsible for these stripes to their offspring.

- A** 1, 2, 3 and 4
- B** 1, 2 and 4 only
- C** 1, 3 and 4 only
- D** 2 and 3 only

- 26** Which does **not** contribute to a shift in allele frequencies in populations?

- A** acquisition of new traits in an organism's lifetime
- B** change in selection pressure from the environment
- C** differential reproductive success
- D** stabilising selection

27 Sarawak is an area of South-East Asia that is largely covered by tropical rainforest. Logging has been allowed in large parts of the forest. A study was carried out to estimate the population size of different species of mammals living in the rainforest:

- before logging
- immediately after logging
- two years after logging
- four years after logging

The table shows the results of the study for six species of mammals. Where numbers were too small to measure the population density, the species were recorded as “present”.

	mean number of animals km ⁻²			
mammal	before logging	immediately after logging	two years after logging	four years after logging
marbled cat	present	0	0	0
small-clawed otter	present	0	0	0
giant squirrel	5	1	4	1
treeshrew	10	5	10	38
small squirrel	16	24	104	19
barking deer	3	1	10	present

Which statements are possible reasons why marbled cats and small-clawed otters are no longer found in this area?

- 1 Population numbers of marbled cats and small-clawed otters were small prior to logging.
- 2 Marbled cats and small-clawed otters are carnivorous and the numbers of prey fell.
- 3 Marbled cats and small-clawed otters were more likely to migrate to other areas.

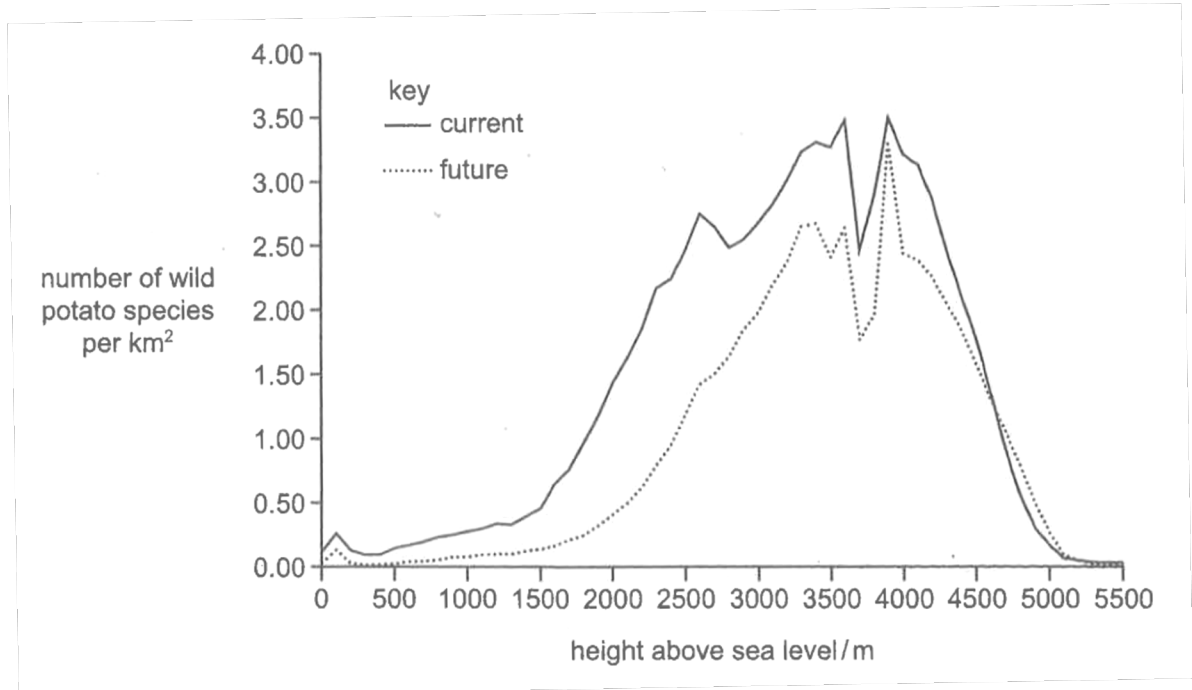
- A** 1, 2 and 3
- B** 1 and 3 only
- C** 2 and 3 only
- D** 2 only

- 28** The table shows the relative proportions of plasma and memory B cells circulating in the blood of a person during a secondary immune response. The relative proportions are given at four different times, A, B, C and D.

At which time will the total number of circulating B lymphocytes be greatest?

	relative proportions	
	plasma cells	memory B cells
A	0	5
B	1	4
C	3	2
D	4	1

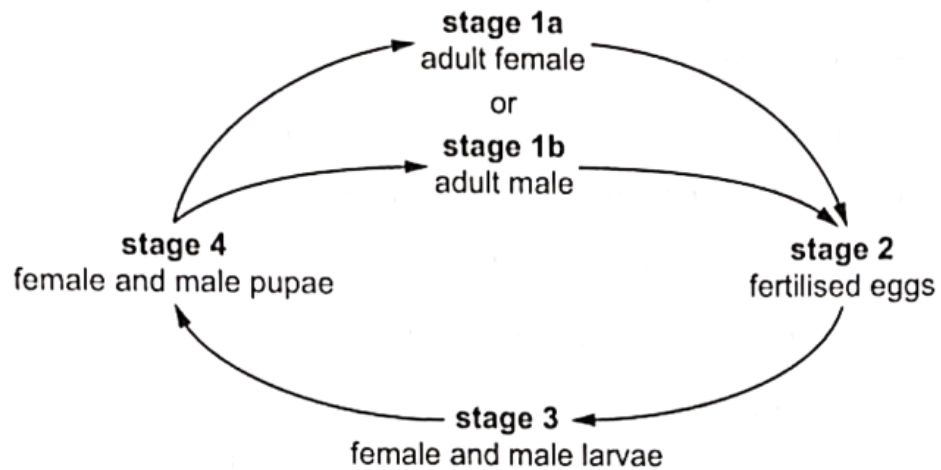
- 29 The ecological niche of over one hundred wild potato species have been analysed in their natural habitats. Projections have been made about the future of these species if global warming occurs. One type of projection is shown on the graph.



Which conclusions about the future effects of climate change can be drawn directly from this graph?

- 1 Some wild potato species may become extinct.
 - 2 The distribution of wild potato species may shift north.
 - 3 The distribution of some wild potato species may shift to a greater height above sea level.
 - 4 There will be less diversity of wild potato species at 5000 m above sea level.
- A** 1 and 3
B 1 only
C 2 and 4
D 2 only

30 The diagram shows the life-cycle of the *Aedes aegypti* mosquito.



At which of the stages of the life-cycle can the mosquito become infected with the dengue disease virus?

- A 1a, 1b, 3 and 4
- B 1a and 1b only
- C 1a only
- D 1b, 3 and 4 only

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